SCOREBOARD – Hadamard 14012 Paley-I (2026-10-08)

Cutoff: 2026-10-08T05:32:03Z (~1:32AM ET)

Object	Status	max|HHˆT-nI|	Wall (s)	SPK
H_14012 Paley-I (q=14011)	PASS	0	2403.7887	**IN SPK** (Paley-I)
H_14012 Paley-II (q=7005)	FAIL/SKIP	–	–	q=7005=3*5*467 NOT prime; ===1 mod 4 but composite
H_14012 Kronecker (4*3503)	FAIL/SKIP	–	–	3503===3 mod 4 NOT Hadamard-admissible; 3503=31*113 (odd composite)

Units: 636/636 * row blocks 110/110 * overall_pass=True * wall_total=2441.0207s
Gap 2052: **OPEN** * clay_closed=false * GATE-04 **LOCKED** * conjecture **OPEN**
IN SPK via Paley-I; Miyamoto digression skipped; exclude Miyamoto-only 8948/8644/8516/...; prior SPK 10224+10212+10164+10160+10152+10140+10112+10104+10100+10068+10040+10008+9968+9932+9924+9908+9888+9884 retained

Next optional: **q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3ˆ2*71317 NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59ˆ2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2ˆ2*59ˆ2 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2ˆ2*3*1163 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3ˆ2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2ˆ2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7001->n=14004 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*1967 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3*2389 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2ˆ2*3*2389 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7013->n=14028 Paley-II DIRECT (FIRST n===0 mod 4 after 14012 with any direct classical Paley route; 14016/14020/14024 have none; q2=7013 prime ===1 mod 4; Paley-I q=14027=13ˆ2*83 NOT prime -> FAIL; Kronecker 14028=4*3507; 3507=3*7167 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3507); 14028=2ˆ2*3*7167 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | Kronecker-only (no direct Paley route; note-only, needs a Kronecker wave, NOT on box): 14032=4*3508 (3508=2ˆ2*877 ===0 mod 4; H_3508 via Paley-II q=1753 prime ===1 mod 4) and 14048=4*3512 (3512=2ˆ3*439 ===0 mod 4; H_3512 via Paley-I q=3511 prime ===3 mod 4) | next Paley-I DIRECT: q=14051->n=14052 (prime ===3 mod 4; 14029/14033 primes ===1 mod 4 skipped; no prime ===3 mod 4 in 14012..14050; Paley-II q=7025=5ˆ2*281 NOT prime**

(===1 mod 4 but composite) -> FAIL/SKIP; Kronecker 14052=4*3513; 3513=3*1171 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3513); 14052=2^2*3*1171 ===4 mod 16 = odd-m shape with m===1 mod 4 (13924/13956/14004-like residues)) (or gap 2052 / Track B)